

Conversa.AI: A Platform for AI Interviewing and Authentic Conversation

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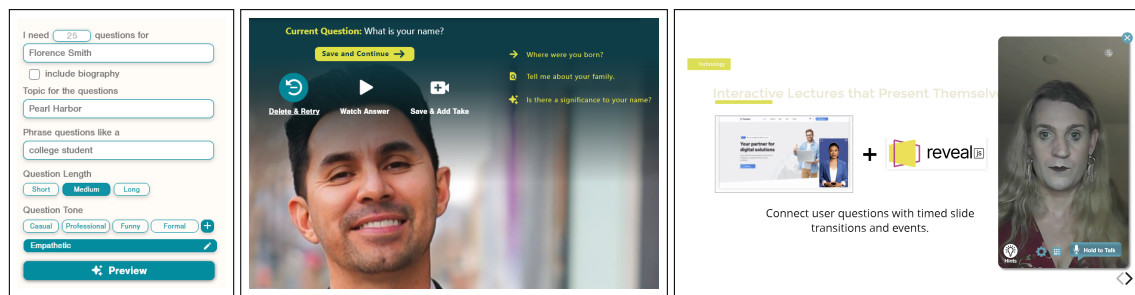


Figure 1: (left) UI for generating questions. (center) Interview UI with suggested follow-up questions (right). Slide deck with embedded interactive video player.

CCS CONCEPTS

- Computing methodologies → Natural language processing.

KEYWORDS

Video interviewing, Natural language processing, Large language models

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1 INTRODUCTION

Communication plays a pivotal role in enhancing productivity and fostering effective collaboration. Not all users are natural storytellers. Even experts internalize knowledge through years of practice and automation, which makes it harder for them to convey

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their underlying processes to novices. Conversational AI is rapidly changing how the world searches for human knowledge. Chat-bots based on large-language models such as ChatGPT or Gemini can quickly summarize public knowledge with minimal training and prompting. These models, however, may exhibit bias and inaccuracies [Brown et al. 2020]. Furthermore, generic LLM models struggle to capture real-world first-person experience. In contrast, traditional intent-based search-and-retrieval chat-bots can provide predictable results, but require significant manual training [Traum et al. 2015]. We address this problem with a set of conversational AI tools to author and publish reusable interactive video interviews. Expert interviews play a key role in knowledge transfer for many real-world applications allowing experts to unpack and preserve their knowledge.

Conversa.AI is a SaaS platform that combines AI dialog management, video authoring, and streaming in one convenient location. We use AI-driven interviewing to gather human expertise with real-time guidance. We then publish interactive videos and presentations with a natural language interface. One interactive video interview can simulate face-to-face dialog with many different audiences simultaneously.

2 INTERVIEW PROCESS AND FEEDBACK

Our system serves as an AI interviewing assistant, automatically generating questions before and during the interview, as well as transcribing the answers in real-time for further analysis as needed.

To prepare for the interview, users provide some contextual information, including the interviewee name, bio, topics of interest, and target user audience. We then automatically create LLM prompts, for example: “You are a high school student learning about WWII for the first time. You’re speaking with a WWII vet named Vincent Speranza. Write 30 questions you would ask him about his war service, personal life and career.” Users can also select questions from a library of over 2000 preexisting questions and topics. During video recording, interviewees record answers using a connected web camera or by uploading existing media. Interviewee responses are immediately uploaded and provide additional context for generative follow-up questions. Follow-up questions also help improve the overall quality of user responses. We compare the transcript with predefined talking points to identify any missing information. We analyze response transcripts for tone and filler words, and can suggest alternative phrasings if needed. Given limited interview time, it is necessary to prioritize potential lines of questioning. The interviewer can flag specific questions as important, while our system identifies questions that may have already been answered by an earlier response. Our AI estimates time for remaining questions as a weighted average of response times to similar historical questions. Different people have varying speech styles so we scale the average based on each subject’s relative speech rate and verbosity.

3 INTERACTIVE VIDEO PLAYER

Our interactive video player simulates a continuous video conversation, connecting video responses with neutral listening videos. Users can ask verbal or text questions or select from a list of hints. Our dialog service uses semantic encoding to match user questions to videos. We run multiple sentence embeddings [Conneau et al. 2020][Yang et al. 2020] in parallel to identify matches in both video transcripts and alternative questions. If a video is unable to answer a question, we use Retrieval Augmented Generation (RAG) to summarize existing documents and thus fill in potential knowledge gaps. Users can interact with videos on a wide range of devices via web browser or within multiple video game engines (Unity, Unreal, PlayCanvas).

4 CONCLUSION AND FUTURE WORK

We have utilized Conversa.AI to for multiple domains, including recording veterans, Holocaust survivors, industry leaders, and family legacies. We are deploying the system for active learning in classrooms, combining interactive Q&A with 3D animations, slide shows and other content. Students retain more information through active engagement and by asking questions in their own words. Future improvements include recognizing interviewee emotional states through language and non-verbal cues, and building trust and empathy over multiple interactions. We will continue expand our community of interactive digital experts and storytellers for both entertainment and education.

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